USN | | | | | | | | | | [22BS2MA03]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

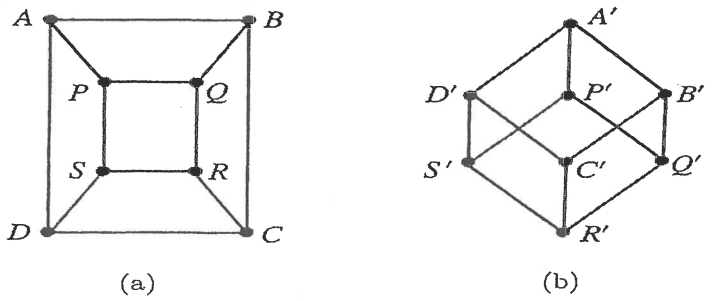
DISCRETE MATHEMATICS AND GRAPH THEORY

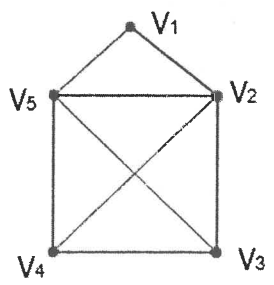
Time: 3 Hours **Maximum Marks: 100**

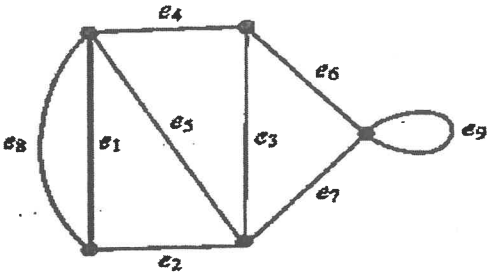
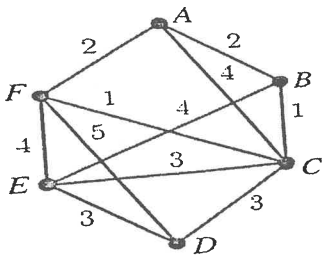
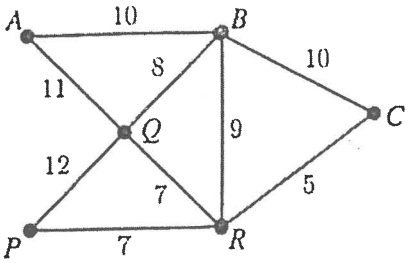
Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART- A	**05 X 02=10**		
1.	**a.**	Cars of a particular manufacturer come in 4 models, 12 colours, 3 engine sizes, and 2 transmission types. **i.** How many distinct cars (of this company) can be manufactured? **ii.** Of these how many have the same colour?	L-2	CO2	**02**
	b.	Find the exponential generating function for the sequence $0!, 1!, 2!, 3!, \ldots$	L-3	CO3	**02**
	c.	Find the general solution of the recurrence relation $a_n = 7a_{n-1} - 10a_{n-2}$	L-3	CO4	**02**
	d.	Define simple graph and multi graph.	L-1	CO1	**02**
	e.	Consider the prefix codes a : 111, b : 0, c : 1100, d : 1101, e : 10 using these codes decode the following sequences **i.** 1001111101 1101111110010	L-2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Find the number of positive integer solutions of the equation $x_1 + x_2 + x_3 = 17$ we have $x_1 \geq 1,\ x_2 \geq 1,\ x_3 \geq 1$.	L-3	CO2	**05**
		OR			
	b.	How many friends must you have to guarantee that at least 5 of them will have birthdays in the same month?	L-3	CO2	**05**
3.	**a.**	Determine the coefficient of x^{18} for $(x + x^2 + x^3 + x^4 + x^5)(x^2 + x^3 + x^4 + \cdots)^5$	L-3	CO3	**05**
		OR			
	b.	Determine the coefficient of x^{20} in $(x^2 + x^3 + x^4 + x^5 + x^6)^5$	L-3	CO3	**05**
4.	**a.**	Solve the recurrence relation $a_n = 7a_{n-1}$, where $n \geq 1$, given $a_2 = 98$.	L-3	CO4	**05**
		OR			

	b.	Suppose that there are $n \geq 2$ person sat a party and that each of these persons shakes hands (exactly once) with all of the other persons present. Using a recurrence relation find the number of hand shakes.	L-3	CO4	**05**
5.	**a.**	A connected planar graph G has 9 vertices with degrees 2, 2, 3, 3, 3, 4, 5, 6, 6. Find the number of regions of G.	L-3	CO5	**05**
		OR			
	b.	Verify that the two graphs shown below are isomorphic. 	L-3	CO5	**05**
6.	**a.**	Exhibit the following: **i.** A graph which has both an Euler circuit and a Hamilton cycle **ii.** A graph which has an Euler circuit but no Hamilton cycle. **iii.** A graph which has a Hamilton cycle but no Euler circuit. **iv.** A graph which has neither a Hamilton cycle nor an Euler circuit.	L-3	CO5	**05**
		OR			
	b.	If a tree T has 3 vertices of degree 2, 2 vertices of degree 3 and 2 vertices of degree 4. Find the number of pendant vertices in T.	L-3	CO6	**05**
7.	**a.**	How many permutations of 1, 2, 3, 4, 5, 6, 7, 8 are not derangements?	L-3	CO2	**05**
		OR			
	b.	Prove that the number of partitions of a positive integer n into distinct summands is equal to the number of partitions of n into odd summands.	L-3	CO3	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Determine the number of positive integer n such that $1 \leq n \leq 100$ and n is not divisible by $2, 3,$ or 5.	L-3	CO2	**10**
		OR			

	b.	An apple, a banana, a mango and an orange are to be distributed to four boys B1, B2, B3, B4 The boys B1 and B2 do not wish to have apple, the boy B3 does not want banana or mango, and B4 refuses orange. In how many ways the distribution can be made so that no boy is displeased?	L-3	CO2	**10**
9.	**a.**	For each of the words given below, find the exponential generating function for the number of ways to arrange ns 11 letters selected from the word. **i.** MISSISSIPPI **ii.** ISOMORPHISM **iii.** MATHEMATICS **iv.** ENGINEERING	L-3	CO3	**10**
		OR			
	b.	Using generating function, find the number of partitions of $n=6$. and list all possible partitions of n=6.	L-3	CO3	**10**
10.	**a.**	Solve the recurrence relation $a_{n+2} - 10a_{n+1} + 21a_n = 3n^2 - 2,\ n \geq 0.$	L-3	CO4	**10**
		OR			
	b.	Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with the initial conditions $a_0 = 1$ and $a_1 = 4$.	L-3	CO4	**10**
11.	**a.**	Find the chromatic polynomial for the graph shown below using decomposition theorem. If 5 colors are available, in how many ways can the vertices of this graph be properly colored?  	L-3	CO5	**10**
		OR			

	b.	Carry out the elementary reduction process for the following graph shown in below Figure and check the planarity. 	L-3	CO5	**10**
12.	**a.**	Explain the following with an example. **i.** Walk **ii.** Trail **iii.** Path **iv.** Circuit **v.** Cycle	L-2	CO6	**10**
		OR			
	b.	Using Prim's algorithm, find a minimal spanning tree of the weighted graph. 	L-3	CO6	**10**
13.	**a.**	Using Kruskal's Algorithm, find a minimal spanning tree for the given weighted graph. 	L-3	CO5	**10**
		OR			
	b.	Construct an optimal prefix code for the message ROAD IS GOOD. Indicate the code.	L-3	CO6	**10**